

Working In Mixed Realities: 3D User Interfaces And Interaction Patterns To Integrate Practices In Different Perceptual Spaces

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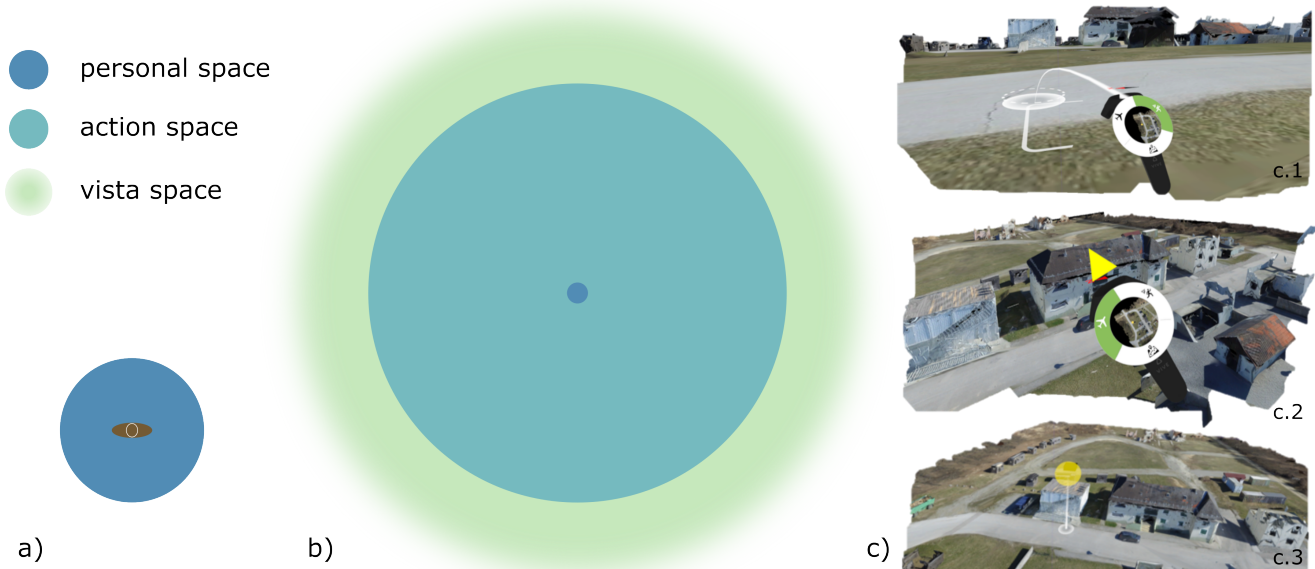


Figure 1: a) Schematic top-down view of a person in their personal space, b) proportion between personal, action and vista space - the vista space is partially represented, since it has no ending limit, c) different navigation interaction patterns from [18] in different perceptual spaces: c.1) teleport, c.2) flying, c.3) puppet placement in world-in-miniature

ABSTRACT

This PhD project proposes a foundational research to classify and analyse how 3d user interfaces and interaction patterns in mixed realities can be integrated to and improve existing working practices according to their perceptual scale. Besides the formulation of a theoretical background, the proposed research will discuss the design challenges of three case studies. In each case the focus of the working activity is on a different scale: personal space, action space and vista space. The final aim is to understand the similarities and the differences between the different projects and provide design guidelines for the design and implementation of 3d user interfaces and interactions patterns in mixed realities according to the main perceptual space of the work activity.

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3651130>

CCS CONCEPTS

- **Human-centered computing** → *Interface design prototyping*;
- **Computing methodologies** → **Mixed / augmented reality**;
Virtual reality; **Perception**.

KEYWORDS

Virtual Reality, Augmented Reality, Extended Reality, Manipulation Techniques, Navigation Techniques

ACM Reference Format:

Marina Lima Medeiros. 2024. Working In Mixed Realities: 3D User Interfaces And Interaction Patterns To Integrate Practices In Different Perceptual Spaces. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3613905.3651130>

1 INTRODUCTION

Recent developments in computer vision and the commercial availability of affordable head-worn displays (HWD) have sparked interest in the use of new technologies for work activities, in addition to their original purpose for gaming. Some systems relate to Extended

Reality (XR), an emerging hybrid technology that covers a spectrum of applications related to Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), which together form the so-called “virtuality continuum” [19]. It is now possible to expand the interactive work environment beyond desktop screens, allowing for immersion and interaction in three dimensions. This may lead to new insights or better solutions for old questions, as well as more productive workflows.

Immersive MR systems should enable interactions to occur with six degrees-of-freedom, allowing movement and rotation in three dimensions [11]. These degrees of freedom enable users to interact with the entire space around them, making the entire environment a potential location for user interface, important information and interactive objects. Mixed reality immersive environments can subvert our space-time logic, the proportion between the body of the user and digital objects can change dynamically, through a change of point-of-view or by altering the scale of the objects. For instance, this allows guided-tours through internal parts of a virus, crowded with proteins [2]. Or the immersive visualisation of a network of relationships between genes [21]. Demolished buildings or those still in planning phase can be visualized on top of the existing city [20]. Large structures can be analysed and manipulated as toys, like handling the Planet in your hands to choose a place, and then diving in a detailed city model [10].

These examples showcase the potential of dynamic change of point-of-view and transformations in user/scene proportion. However the best practices and standards for this interactivity do not exist yet. And when it actually adds value to a work routine is also an open question. Different authors have identified and classified the interaction patterns and techniques [11, 12, 22, 25], but most of the examples of navigation, selection, manipulation, and system control interactions were thought of for games. It is a fact that the game industry plays an important role in the development of MR devices and up to today create new interaction techniques to be analysed and classified by researchers of the field [26]. While the narrative of games can induce the players to use a plethora of different and sometimes very complex interaction techniques, for working activities simplicity and low mental effort are key aspects for technology acceptance in workplaces [29]. And the use of MR technologies in workplaces should be combined with changes in organizational practices [23]. Other approaches have considered domain-specific interactions for particular use cases [4] and emerging fields like immersive analytics might need special interaction solutions [8].

Another aspect is that dynamically changing the proportion between viewer and the observed object or information changes the way the object is perceived by the user, because humans do not perceive space equally. There are regions around us with different perception characteristics, known as perceptual spaces [6, 27] (see Figure 1):

- **Personal space:** within 1.5 m to 2 m is the space where we can usually reach and manipulate objects.
- **Action space:** is located from 1m to 30m from a person’s body. In this space is where social interaction usually happens, where it is possible to recognize other’s faces and to talk.

- **Vista space:** is all the space that we can see more than 30m away from our bodies. In these areas objects are seen, but their characteristics and motion are considerably less salient for the human eye.

Perception spaces were defined decades ago and have been fundamental in the development of hardware and software for MR technologies. This allowed the composition of scenes and more effective rendering techniques for HMDs that give the user the illusion of a three-dimensional space and the feeling of being immersed in that space, although it is only represented on flat screens in front of their eyes. In this thesis, we want to use this classification as an analysis category for MR applications for work activities to understand if the perceptual scale in which the information is presented and the transition between them are critical to improve the usefulness of the applications.

The thesis will discuss the design challenges of three case studies, through a foundational research methodology. The aim is to understand the similarities and differences between the projects and provide design guidelines for 3D user interfaces and interaction patterns in mixed realities based on the main perceptual space of the work activity.

2 RELATED WORK

Designing a MR application for specialised work is a complex activity that involves many different combinations of independent variables. To find guidelines to navigate those combinations according to the application’s goal and users preferences, researchers are exploring design spaces, checking possible spatial combinations of interface elements and evaluating the most suitable options. Analysing office ergonomics and the field of view of commercial HWDs, Mike Alger [1] defined a main content zone (from 50 cm to 130 cm) in the front of a sitting user that would be optimal for organising an operational system in VR. The author, however, focuses his work in the placement of menus and considers that the work will be done mostly in a stationary setup.

The transition between 2D and 3D representations of data was the subject of a design space by Lee et al. [13]. The researchers described possible tasks that these transformations can fulfil when transitioning between four main states: single-view 2D, single-view 3D, multi-view 2D, and multi-view 3D. Changes in data over time was the topic of Fouché et al. [9] who proposed a design space for the immersive analysis of spatial data distributed on a 3D timeline. Liu et al. [15] proposed a design space for immersive layouts of small multiples data and analysed the different solutions, concluding that the performance of different layouts is dependent on the number of multiples displayed and pointed out which layout is better according to the number of multiples. More recently they expanded their layout exploration to view management in three dimensions, analysing the combination of different view presentation layouts with different interaction styles and action [14]. It is important to mention that in their work they consider a proximity dimension for view presentation, classifying it as on body, near, or far which we can relate to the perceptual spaces previously mentioned [6, 27]. The authors, however, only focus on the display and management of view, not expanding their work to larger and tridimensional information.

The focus on small scale objects or “information containers” and their spatial distribution are perceived in most of recent work, with interactions with these objects being contained mostly in the personal space and partially in the action space. Chen et al. [5] propose a theoretical model of a design space for urban immersive analytics where they characterised visualisations and proposed a typology for combining 2D and 3D visualisations of urban models that would fit also in vista space. Danyluk et al. [7] explored the design space of Worlds in Miniature (WiM), a interaction pattern that combines manipulation and navigation of objects in different sizes. The authors analysed eight design dimensions, including the Size-Scale-Scope and Reference Frame of WiMs of 25 papers to provide guidelines for future work on the topic. The paper, however, do not evaluate which solutions are better according to each size-scale-scope of the object to be visualized.

Wagner et al. [28] analysed how different exploration modes (navigation or manipulation) and frame of reference (egocentric or exocentric) can influence data analysis in immersive environments. They considered three environments: room-scale exocentric, room-scale egocentric, and large-scale egocentric view. The evaluation suggests that the egocentric room-scale exploration required less mental load from the users, while exocentric exploration improved some task performances. Also the combination of navigation through data and manipulation of the data visualisation reduced workload, temporal demand, and physical effort. This work suggests that the combination of interaction patterns in different perceptual spaces is fundamental to improve work routines in MR as we consider for our research project.

3 RESEARCH QUESTIONS

- Working in different space perceptual scales demands different 3d interfaces and interaction techniques?
- Does transitioning objects between perceptual spaces improve the usefulness of a MR application?

4 SPECIFIC OBJECTIVES

- Analyse and classify 3d user interfaces and interaction patterns that transition from one perceptual space to other and their importance for work applications in MR.
- Provide design guidelines for the design and implementation of 3d user interfaces and interactions patterns in MR according to the perceptual scale of the work activity.

5 RESEARCH METHODS

The thesis will be structured on the foundational research of three different study cases based on projects of the VRVis Research Center, Austria’s leading institution in the field of visual computing with large experience in connecting basic research with the needs of professionals in different industries, which is critical to this thesis. Foundational research is part of User Experience research routine focused on qualitative methods to define the needs of the users. Currently, these methods are underused in computer vision and graphics research but could provide meaningful insights for this field of knowledge[3].

The three mixed reality projects have in common the challenge to integrate MR applications in working practices of different fields,

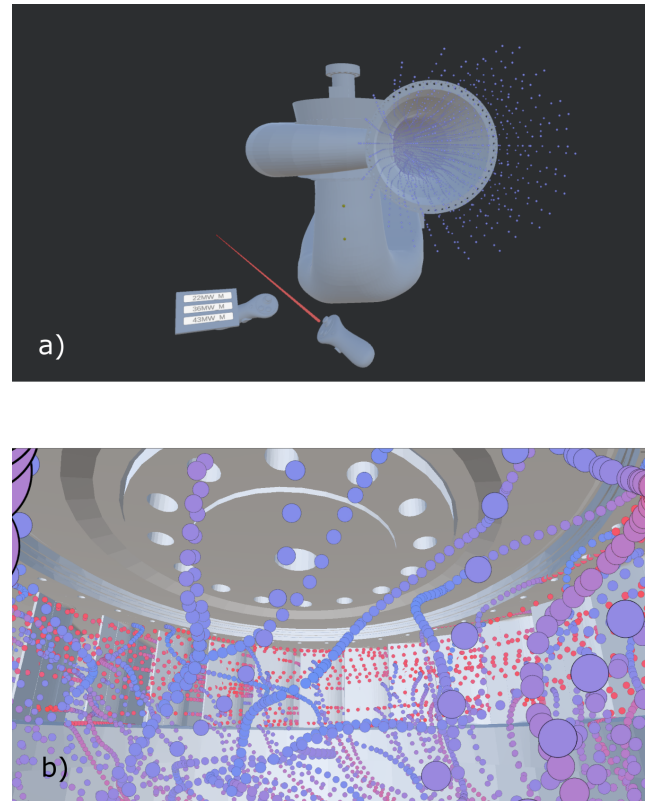


Figure 2: a) General view of the water turbine in VR environment for the DIGI-Hydro project [17], b) internal view of runner with simulated flow data from the DIGI-Hydro project [17]

to be designed in collaboration with specialists of each field and, most importantly, the assignments of each project are related to spatial and abstract data (3d models and visual information) of different sizes that require to be visualised in different perceptual spaces to be seen in real scale.

Here is a short description of the three use cases:

- **Use Case 1:** MilGeoVA is a virtual reality application for mission planning through terrain exploration, mapping and annotation for the Austrian Army. 3D models elaborated with photogrammetric data are loaded and explored in VR through different points of view, the user can explore and annotate the terrain to create a tactical solution. A real scale representation of the models to be visualised places most of the models in the vista space of the viewer [18] (see Figure 1c).
- **Use Case 2:** DIGI-Hydro combines and compares simulated and sensor-measured data of hydroelectric power plants to enable data-driven predictions of the lifetime of mechanical parts. Most of the work and data visualisation was done in desktop applications but a Virtual Reality application for the immersive visualisation and analysis of the data in the context of the 3d model of the water turbine is under development. The project is done in partnership with the Institute

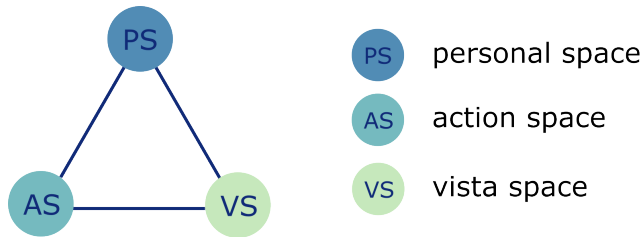


Figure 3: Personal, action and vista space organized as a triangle, inspired in the analytical lenses of [24]

of Energy Systems and Thermodynamics at the Vienna University of Technology and the hydro turbine visualised in this project fits in the action space of the user [17] (see Figure 2a-b).

- **Use Case 3:** ClimaSens is a project of urban micro-climate modelling with interactive, collaborative, and immersive planning workflows that aims to have accessible solutions for scientists, professionals and also citizens in general. The project deals mostly with data visualisation graphics and dashboard solutions that fit the personal space of the viewer, some simulation results might be visualised in action and vista space as well, depending on the area considered for modelling.

The projects will be analysed according to definitions and concepts of a theoretical background. A strong qualitative assessment of the design process will be made through the documentation of the process, needfinding techniques, interviews with users and quantitative evaluations. The projects will be compared to check for similarities and differences in the 3d user interface and interaction patterns according to the perceptual spaces that are part of the user experience. We are also considering creating a graphical representation of the transitions between perceptual spaces done by users while performing a task, inspired by [24] (see Figure 3).

6 RESULTS AND CONTRIBUTIONS TO DATE

TU Vienna Informatics Doctoral School procedure consists of five phases: Enrollment, Proficiency Evaluation, Pre-Submission, Official Submission and Defense. Enrollment consists of the application procedure, after that the Ph.D. student can enrol in fundamental and area courses and start a dissertation proposal that will be presented in the Proficiency Evaluation. If the student is approved in the Proficiency Evaluation, they can start writing the dissertation that will be submitted in the subsequent phases. The TU Vienna Informatics Doctoral School also requires that the doctoral candidate publishes at least three papers in high quality conferences or five papers in journals as first author that will be used as the basis of the dissertation. This research proposal is currently entering in the Proficiency Evaluation phase and also has produced some publications:

- **ISMAR Workshop Paper [16]** This short paper contains the first steps of the methodology used in this thesis. A classification of the interaction patterns and techniques according to Jerald [11] and an analysis of which patterns and techniques were used for different perceptual scales and for the

transition between them for **use case 1**. The plan is to improve this methodology and reapply in the other two use cases.

- **IEEEVR Paper [18]** This paper presents the foundational research leading to the development of the application of **use case 1**. The qualitative evaluation shows that immersive views of spatial data are beneficial for understanding geometric structures, which is mainly achieved by the user's transition between perceptual spaces.
- **SUI Poster [17]** This poster presents the first steps of the foundational research and first implementations of **use case 2** project.

7 OPEN QUESTIONS AND NEXT STEPS

For future work, we will continue to focus on the second use case project by reproducing the analysis of the interaction patterns and techniques used in the first use case, and further evaluating the application.

The third use case project is still in the planning phase. This doctoral consortium presents a great opportunity to add improvements to the foundational research process of this project.

The research questions remain partially open as the last two use cases have not yet been analysed and evaluated and the first use case can also go through a new evaluation. The CHI '24 Doctoral Consortium provides an excellent opportunity to enhance the proposal before it is submitted to the Proficiency Evaluation.

8 DISSERTATION STATUS AND LONG-TERM GOALS

This thesis is supervised by Professor Hannes Kaufmann from the Virtual and Augmented Reality research unit of the Institute of Visual Computing and Human-Centered Technology of the Vienna University of Technology. The dissertation writing process will commence after passing the Proficiency Evaluation. I will complete the Proficiency Evaluation during the second semester of this year, submit a paper for publication on the second use case also during the second semester. I will work on the paper for the third use case project next year, and complete the dissertation by the end of 2025.

ACKNOWLEDGMENTS

VRVis is funded by BMK, BMDW, Styria, SFG, Tyrol, and Vienna Business Agency in the scope of COMET - Competence Centers for Excellent Technologies (879730) which is managed by FFG.

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